

Question 1 – Scatterplot Analysis

R Code

```
# Question 1: Scatterplot Analysis
```

```
Study_Hours <- c(2,3,4,5,6,7,8,9,10,11,12,13)
```

```
Exam_Score <- c(45,50,55,61,64,68,72,76,81,84,88,92)
```

```
# Create data frame
```

```
data <- data.frame(Study_Hours, Exam_Score)
```

```
# Scatterplot
```

```
plot(data$Study_Hours, data$Exam_Score,  
     main = "Study Hours vs Examination Score",  
     xlab = "Study Hours",  
     ylab = "Exam Score",  
     pch = 19)
```

```
# Fit regression model
```

```
model <- lm(Exam_Score ~ Study_Hours, data = data)
```

```
# Add regression line
```

```
abline(model, lwd = 2)
```

```
# Display model summary
```

```
summary(model)
```

Interpretation

- The scatterplot shows a **strong positive relationship**.
- As study hours increase, examination scores also increase.

- The fitted regression line confirms the upward trend.
- Therefore, students who study more hours tend to obtain higher examination scores.

Question 2 – Scatterplot with Grouping

Since the question does not provide the 15 observations, the following sample data contains **5 observations for each region**.

R Code

Question 2: Grouped Scatterplot

```
Advertising_Expenditure <- c(
  10, 15, 20, 25, 30,    # North
  12, 18, 23, 28, 35,    # South
  8, 14, 19, 24, 29     # West
)
```

```
Sales <- c(
  50, 58, 67, 75, 84,    # North
  48, 60, 69, 80, 91,    # South
  42, 51, 60, 68, 77     # West
)
```

```
Region <- c(
  "North", "North", "North", "North", "North",
  "South", "South", "South", "South", "South",
  "West", "West", "West", "West", "West"
)
```

```
data2 <- data.frame(
```

```
Advertising_Expenditure,  
Sales,  
Region  
)
```

```
# Different symbols for each region
```

```
symbols <- c("North" = 19, "South" = 17, "West" = 15)
```

```
# Different plotting symbols
```

```
plot(data2$Advertising_Expenditure[data2$Region == "North"],  
     data2$Sales[data2$Region == "North"],  
     pch = symbols["North"],  
     xlim = range(data2$Advertising_Expenditure),  
     ylim = range(data2$Sales),  
     xlab = "Advertising Expenditure",  
     ylab = "Sales",  
     main = "Advertising Expenditure vs Sales by Region")
```

```
points(data2$Advertising_Expenditure[data2$Region == "South"],  
       data2$Sales[data2$Region == "South"],  
       pch = symbols["South"])
```

```
points(data2$Advertising_Expenditure[data2$Region == "West"],  
       data2$Sales[data2$Region == "West"],  
       pch = symbols["West"])
```

```
# Legend
```

```
legend("topleft",
```

```
legend = c("North", "South", "West"),
```

```
pch = c(19, 17, 15))
```

Interpretation

- All three regions show a **positive relationship** between advertising expenditure and sales.
 - As advertising expenditure increases, sales generally increase.
 - The relationship appears reasonably strong in all three regions.
 - South shows slightly higher sales at higher advertising expenditure levels.
-

Question 3 – Correlation and Correlogram

R Code

```
# Question 3: Correlation and Correlogram
```

```
Temperature <- c(22,24,25,27,29,30,31,33,34,36)
```

```
Humidity <- c(82,78,75,70,68,65,63,58,55,52)
```

```
Pressure <- c(1015,1013,1012,1010,1008,1007,1005,1003,1002,1000)
```

```
Wind_Speed <- c(8,9,10,12,11,14,13,15,16,17)
```

```
Rainfall <- c(18,16,14,12,11,9,8,6,5,3)
```

```
weather <- data.frame(
```

```
  Temperature,
```

```
  Humidity,
```

```
  Pressure,
```

```
  Wind_Speed,
```

```
  Rainfall
```

```
)
```

```
# Correlation matrix
```

```
cor_matrix <- cor(weather)
```

```
print(cor_matrix)
```

```
# Correlogram using base R
```

```
image(1:ncol(cor_matrix),  
      1:ncol(cor_matrix),  
      cor_matrix,  
      axes = FALSE,  
      col = colorRampPalette(c("blue", "white", "red"))(100),  
      xlab = "",  
      ylab = "",  
      main = "Correlogram - Weather Variables")
```

```
# Add axes
```

```
axis(1, at = 1:ncol(cor_matrix),  
     labels = colnames(cor_matrix),  
     las = 2)
```

```
axis(2, at = 1:ncol(cor_matrix),  
     labels = colnames(cor_matrix),  
     las = 2)
```

```
# Add correlation values
```

```
for(i in 1:nrow(cor_matrix)) {  
  for(j in 1:ncol(cor_matrix)) {  
    text(i, j, round(cor_matrix[i, j], 2))  
  }  
}
```

}

Interpretation

The correlogram uses:

- **Red** → positive correlation
- **Blue** → negative correlation
- **White** → weak or near-zero correlation

The numbers inside the cells show the actual correlation coefficient.

For example:

- Temperature and Humidity show a strong negative relationship.
- Temperature and Wind Speed show a positive relationship.
- Temperature and Pressure show a negative relationship.

Question 4 – Correlogram for Multivariate Data

R Code

```
# Question 4: University Student Data
```

```
Attendance <- c(60,65,70,72,75,78,80,83,86,90,92,95)
```

```
Assignment_Score <- c(  
  55,60,64,68,70,73,76,79,82,86,89,93  
)
```

```
Internal_Marks <- c(  
  50,55,61,65,68,71,74,78,81,85,88,92  
)
```

```
Study_Hours <- c(  
  2,3,4,4,5,5,6,6,7,8,9,10
```

```
)
```

```
Final_Marks <- c(  
  52,58,63,67,70,74,77,81,84,88,91,95  
)
```

```
student_data <- data.frame(  
  Attendance,  
  Assignment_Score,  
  Internal_Marks,  
  Study_Hours,  
  Final_Marks  
)
```

```
# Correlation matrix  
cor_matrix4 <- cor(student_data)
```

```
print(cor_matrix4)
```

```
# Correlogram  
image(1:ncol(cor_matrix4),  
      1:ncol(cor_matrix4),  
      cor_matrix4,  
      axes = FALSE,  
      col = colorRampPalette(c("blue", "white", "red"))(100),  
      xlab = "",  
      ylab = "",  
      main = "Correlogram - Student Performance")
```

```

# X-axis
axis(1,
     at = 1:ncol(cor_matrix4),
     labels = colnames(cor_matrix4),
     las = 2)

# Y-axis
axis(2,
     at = 1:ncol(cor_matrix4),
     labels = colnames(cor_matrix4),
     las = 2)

# Correlation values
for(i in 1:nrow(cor_matrix4)) {
  for(j in 1:ncol(cor_matrix4)) {
    text(i, j, round(cor_matrix4[i, j], 2))
  }
}

```

Interpretation

The correlogram shows the strength and direction of relationships among the five variables.

Based on the sample data:

- **Assignment Score** has a very strong positive relationship with Final Marks.
- **Internal Marks** also have a strong positive relationship with Final Marks.
- **Attendance** has a positive relationship with Final Marks.
- **Study Hours** has a positive relationship with Final Marks.

Therefore, **Assignment_Score** appears to have the strongest relationship with **Final_Marks** in this sample.

The exact strongest variable depends on the actual dataset provided by your question. Since no numerical data was included, this conclusion is based on the sample data above.

Question 5 – Integrated Scatterplot and Correlogram Analysis

Again, because the question doesn't provide the 15 numerical observations, the following is a sample dataset.

R Code

```
# Question 5: Retail Dataset
```

```
Price <- c(  
  100,110,120,130,140,  
  150,160,170,180,190,  
  200,210,220,230,240  
)
```

```
Advertising <- c(  
  10,12,15,18,20,  
  22,25,27,30,32,  
  35,38,40,43,45  
)
```

```
Customer_Visits <- c(  
  50,55,60,65,70,  
  75,80,85,90,95,  
  100,105,110,115,120  
)
```

```
Discount <- c(  
  10,12,15,18,20,  
  22,25,27,30,32,  
  35,38,40,43,45  
)
```

```
5,6,7,8,10,  
9,12,11,14,13,  
15,16,17,18,20  
)
```

```
Sales <- c(  
40,45,50,56,60,  
65,72,75,82,88,  
93,100,105,112,120  
)
```

```
retail <- data.frame(  
  Price,  
  Advertising,  
  Customer_Visits,  
  Discount,  
  Sales  
)
```

```
# -----  
# (a) Scatterplots  
# -----
```

```
par(mfrow = c(2, 2))
```

```
plot(retail$Price, retail$Sales,  
     pch = 19,  
     main = "Sales vs Price",
```

```
xlab = "Price",  
ylab = "Sales")
```

```
plot(etail$Advertising, etail$Sales,  
     pch = 19,  
     main = "Sales vs Advertising",  
     xlab = "Advertising",  
     ylab = "Sales")
```

```
plot(etail$Customer_Visits, etail$Sales,  
     pch = 19,  
     main = "Sales vs Customer Visits",  
     xlab = "Customer Visits",  
     ylab = "Sales")
```

```
plot(etail$Discount, etail$Sales,  
     pch = 19,  
     main = "Sales vs Discount",  
     xlab = "Discount",  
     ylab = "Sales")
```

```
# Reset plotting layout  
par(mfrow = c(1, 1))
```

```
# -----
```

```
# (b) Correlation Matrix
```

```
# -----
```

```
cor_matrix5 <- cor(retail)
```

```
print(cor_matrix5)
```

```
# -----
```

```
# (c) Correlogram
```

```
# -----
```

```
image(1:ncol(cor_matrix5),  
      1:ncol(cor_matrix5),  
      cor_matrix5,  
      axes = FALSE,  
      col = colorRampPalette(c("blue", "white", "red"))(100),  
      xlab = "",  
      ylab = "",  
      main = "Correlogram - Retail Variables")
```

```
axis(1,  
     at = 1:ncol(cor_matrix5),  
     labels = colnames(cor_matrix5),  
     las = 2)
```

```
axis(2,  
     at = 1:ncol(cor_matrix5),  
     labels = colnames(cor_matrix5),  
     las = 2)
```

```
# Add correlation values
for(i in 1:nrow(cor_matrix5)) {
  for(j in 1:ncol(cor_matrix5)) {
    text(i, j, round(cor_matrix5[i, j], 2))
  }
}
```

(d) Interpretation

From the scatterplots and correlogram:

- **Customer Visits vs Sales** shows a strong positive relationship.
- **Advertising vs Sales** shows a strong positive relationship.
- **Price vs Sales** also shows a positive relationship in this sample.
- **Discount vs Sales** shows a positive relationship, but it is weaker than Customer Visits and Advertising.
- Variables with correlation values close to **+1** have strong positive relationships.
- Variables with correlation values close to **-1** have strong negative relationships.
- Variables with correlation values close to **0** have weak linear relationships.

The scatterplots should be compared with the correlation matrix because the correlation coefficient only measures **linear association** and may not reveal nonlinear patterns or outliers.

Quick exam-style answers

Question Main result

- | | |
|-----------|--|
| Q1 | Strong positive relationship between Study Hours and Exam Score |
| Q2 | All regions show positive Advertising–Sales relationships |
| Q3 | Correlation matrix + correlogram identify positive/negative relationships |
| Q4 | Assignment Score has the strongest relationship with Final Marks in the sample |

Question Main result

Q5 Customer Visits and Advertising show strong positive relationships with Sales

Important

For **Q2–Q5**, your question says "the dataset contains..." but **doesn't actually give the numerical data**. If this is an assignment where your teacher expects specific numerical results, replace my sample values with the actual observations from your question. The plotting/correlation code itself can remain essentially the same.